FRI Natural language processing course 2026
Project report

University *of Ljubljana*
Faculty *of Computer and Information Science*

Conversational Recommender System for Vehicles

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Abstract
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Keywords
Keyword1, Keyword2, Keyword3 ...

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Introduction

In the past years, choosing a suitable vehicle has become very challenging due to a wide range of options. Customers can choose between many models with different fuel types, engine characteristics, size, and other specifics.

Most existing systems that assist users in selecting vehicles are based on predefined filters, such as price range, brand, transmission types, and others. While this approach is effective for experienced users, it might not be that suitable for those who cannot easily translate their needs into technical details.

As an improvement to such systems, Conversational Recommender Systems (CRS) have been introduced. A CRS is an interactive system that allows users to express their preferences through dialogue, while the system can ask follow-up questions to refine these preferences and provide the best recommendation(s).

In this paper, we propose a CRS for vehicle selection using retrieval-augmented generation (RAG), with which relevant vehicle data is retrieved and provided to a large language model (LLM) that generates personalized recommendations. The final product is a system that simplifies making a decision for non-expert users, and also improves customer experience in vehicle selling companies.

Related work

Several studies have explored recommender systems, however the vehicle domain remains limited. We categorized existing work into recommendation systems for vehicles and conversational recommender systems for various products.

In the context of vehicle recommendation, Beddouda et al. [1] proposed a system using BERT4Rec, which predicts which cars a user might be interested in based on past interactions (without any user communication). Singh et al. [2] introduced a one-turn dialogue system that uses LDA model to analyse user reviews, where users describe their car preferences and the system returns top few recommendations.

For general CRS, several multi-turn dialogue systems have been proposed, especially in the travel domain. Liao et al. [3] proposed a deep recommender system using seq2seq with a neural latent topic component. Baizal et al. [4] introduced an ontology-based system, and Carrillo et al. [5] introduced a system that uses Chatbot to collect user preferences and generates similarity-based recommendations using SBERT embeddings.

In addition to the travel domain, CRS have been implemented in several other domains. Sun et al. [6] developed a reinforcement learning based CRS for restaurant recommendation using Yelp review data. Zhu et al. [7] proposed a movie recommendation CRS using retrieval-augmented generation (RAG) approach in combination with a large language model (LLM).

Methods

The proposed system is designed as a Conversational Recommender System (CRS) that integrates large language models (LLMs), structured data processing, and retrieval-augmented generation (RAG) to provide personalized vehicle recommendations through natural language interaction. The method begins with a natural language interface through which users describe their preferences and requirements. These inputs are processed by an LLM to extract structured representations of user intent, such as budget constraints, vehicle type, and qualitative preferences.

The system then employs a dual data strategy. A structured database containing quantitative vehicle attributes (e.g., price, fuel consumption, engine specifications) is used to perform constraint-based filtering and reduce the candidate set. In parallel, a vector database built from unstructured sources from brochures and catalogs enables semantic retrieval of relevant contextual information. We will use Auto Catalog Archive [8] as our initial dataset. This website contains more than 55,000 car brochures that include many different car models. Each brochure provides technical specifications (like power, weight, range. . .), list of features (like type of wheel, roof rail, safety equipment. . .), and other promotional information.

Evaluation of a model

Evaluation ideas:

Test if RAG is working well

Build a small controlled dataset (10 very different cars), where only one is meeting the requirements. Ask the system to recommend a car and check if it picks only the valid one.

Compare CRS to other LLMs

Another idea is also creating a few prompts, for which we find a set of suitable cars. Then we compare the output of our CRS system and one another LLM (preferably one that doesn't have access to web). Then we evaluate if the LLMs found any car from the set.

Compare CRS to some website

Compare recommendation to some website (Carwow, Autotrader according to chatgpt) and compare for the same requirements the models output, though it is subjective method of evaluation (hard to tell which model is better, no hard truth on this topic)

Advanced method - baseline filter model

Ideally (but too complicated to implement in such short time) would be to create a basic filtering system, and conmpare the outputs with what the LLM is offering. Again the idea wouldn't be to show it is better output objectively but either it shows more option, or wilder choices, etc. Especially if the requirements are not precise or not coherent between themselves (for instance big car + low consumption + high acceleration)

Equations

Lists

Figures

Tables

Code examples

Results

Discussion

Use the Discussion section to objectively evaluate your work, do not just put praise on everything you did, be critical and exposes flaws and weaknesses of your solution. You can also explain what you would do differently if you would be able to start again and what upgrades could be done on the project in the future.

Acknowledgments

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